

TCS CORPORATE FINANCIAL INTELLIGENCE & FORECASTING

Financial Statement Analytics | Category Intelligence | Financial Health | Forecasting

ANALYTICAL PERIOD	PRIMARY PLATFORM	DELIVERABLE
2004–2026 Historical 2027–2030 Forecast	Python / Google Colab SQL / Power BI / DAX	Interactive Executive Dashboard with Analytical Report

Prepared for Academic / Portfolio Submission

This report documents the end-to-end analytical workflow used to transform the supplied TCS financial dataset into a validated Power BI decision-support dashboard.

Data integrity note: the calculations and conclusions in this report are based on the project dataset and analytical outputs produced during the workflow. Forecasts and the Financial Health Score are analytical constructs for this project and should not be interpreted as official TCS guidance or an investment rating.

1. Executive Summary

This project develops an end-to-end financial intelligence and forecasting solution using the supplied Tata Consultancy Services (TCS) financial dataset. The workflow covers data preparation, standardisation, duplicate verification, descriptive analysis, category classification, SQL-oriented analysis, forecasting and Power BI dashboard development.

The final dashboard provides a single executive view of historical financial movement, financial-category composition, a project-derived financial health indicator, and forecasted Reserves and Surplus for 2027–2030. Interactive Year and Category slicers allow users to investigate the analytical data without changing the underlying source dataset.

Headline dashboard results

KPI	Dashboard Result
Total Financial Amount	3.51M
Average Financial Amount	8.98K
Financial Health Score	50.08 / 100
Historical Years	23 (2004–2026)
Forecasted Reserve	132.24K

The historical trend shows a clear long-term increase in aggregate financial amount, while the category analysis indicates that Equity is the dominant component in the project's aggregated category view. The forecast visualization shows continued growth in the projected Reserves and Surplus series through 2030.

2. Business Problem and Objectives

Financial statement datasets often contain inconsistent year labels, mixed numeric formats, repeated observations, heterogeneous indicator names and a structure that is difficult to consume directly in analytical tools. The project addresses this problem by converting the supplied financial statement data into an analysis-ready structure and then building a management-oriented dashboard.

Project objectives

- Standardise reporting-year labels into a consistent four-digit year structure.
- Convert financial amounts into a reliable numeric format while preserving the original analytical meaning.
- Verify duplicate records using Indicator and Year as the analytical key.
- Create a clean analytical dataset and classify financial indicators into meaningful categories.
- Generate descriptive statistics and identify the largest cumulative financial components.
- Create a project-derived Financial Health Score for dashboard monitoring.
- Develop a forecast of Reserves and Surplus for 2027–2030.
- Deliver an interactive Power BI dashboard with filters for Year and Category.
- Translate analytical outputs into business insights and recommendations.

3. Data Preparation and Quality Management

Because the dataset represents real financial information rather than dummy data, the preparation stage was treated as a controlled transformation process. The original dataset was retained separately while analytical copies were created for cleaning and modelling.

Observed preparation outputs

Validation item	Observed project result
Original dataset	920 rows × 3 columns
Clean analytical dataset	391 rows × 3 columns
Working analytical dataset	920 rows × 3 columns
Duplicate verification	0 duplicate records for Indicator + Year
Core fields	Indicator, Year, Amount

Year standardisation: reporting labels were converted into four-digit calendar years. The final historical dashboard period is 2004–2026.

Numeric handling: Amount values were converted to numeric form after removing formatting characters such as comma separators. Conversion was performed on analytical copies so that the original source structure remained protected.

Duplicate control: duplicate records were checked using Indicator + Year. The project validation returned zero duplicate records.

Category classification: cleaned indicators were grouped into Equity, Borrowings, Other, Liabilities and Assets for category-level analysis.

Important: the 391-row clean analytical dataset reflects the project's indicator-selection/cleaning logic. It should not be interpreted as evidence that the original financial statements contained only 391 observations.

4. Exploratory Financial Analysis

Descriptive statistics were generated after converting Amount to numeric form. The working dataset contained 427 numeric observations for the descriptive-statistics output shown during the analysis.

Statistic	Value
Count	427
Mean	209.150375
Standard deviation	259.631402
Minimum	0.00
25th percentile	0.00
Median	101.00
75th percentile	336.18
Maximum	997.13

The descriptive distribution is positively spread, with the mean above the median and a maximum substantially above the median. The presence of zero-valued observations is visible in the lower quartile and minimum and should therefore be considered when interpreting averages and category totals.

Largest cumulative financial components identified in the analysis

Indicator	Cumulative Amount
Minority Interest	9,011.17
Deferred Tax Liabilities [Net]	8,350.33
Intangible Assets	6,423.65
Capital Work-In-Progress	6,035.48
Non-Current Investments	5,794.70
Total Share Capital	5,724.58
Equity Share Capital	5,124.58
Bonus Equity Share Capital	4,274.63
Non-Current Investments Unquoted Book Value	3,843.75
Total Non-Current Liabilities	3,823.75

These values are cumulative across the analyzed observations and are used to identify comparatively large components in the project dataset; they are not single-year balance-sheet values.

5. Financial Category Intelligence

The cleaned financial indicators were classified into five analytical categories: Equity, Borrowings, Other, Liabilities and Assets. This classification enables management-level interpretation without requiring users to inspect hundreds of individual indicator names.

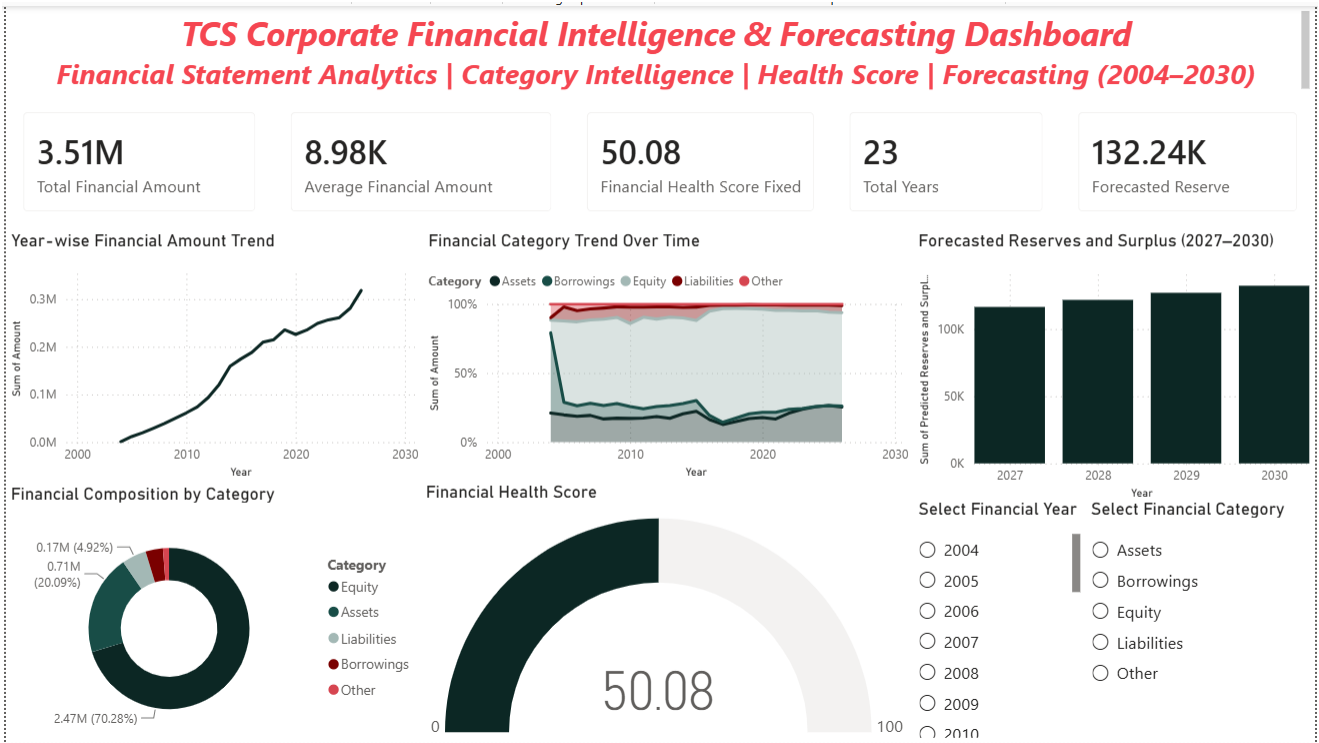
Category	Count of indicators
Equity	115
Borrowings	92
Other	69
Liabilities	69
Assets	46

The dashboard's category composition view shows Equity as the largest aggregate component, accounting for approximately 70.28% of the displayed financial composition. Assets account for approximately 20.09%. The remaining share is distributed among the other analytical categories.

The category trend visual also demonstrates that the relative composition changes across years. This makes the category view useful for identifying structural shifts that may not be obvious from a single total-financial-amount trend.

6. Power BI Executive Dashboard

The final Power BI dashboard combines KPI cards, historical trends, category intelligence, financial health monitoring, forecasting and interactive slicers. The dashboard is designed as an executive summary rather than a replacement for the underlying financial statements.



Interactive controls: Select Financial Year and Select Financial Category.

Dashboard KPI snapshot: Total Financial Amount 3.51M; Average Financial Amount 8.98K; Financial Health Score 50.08; Total Years 23; Forecasted Reserve 132.24K.

7. Financial Health and Forecasting

Financial Health Score. A project-derived Financial Health Score was incorporated into the dashboard as a 0–100 gauge. The final dashboard displays a value of 50.08. This score is a custom analytical indicator created for dashboard monitoring; it is not an official TCS financial rating, credit score or external investment recommendation.

Forecasting. The project includes a dedicated forecast table for Reserves and Surplus covering 2027–2030. The Power BI visualization presents the four forecast years as an upward sequence, with the dashboard KPI showing a combined forecasted reserve of 132.24K.

The forecast should be interpreted as a model-based analytical projection rather than a management forecast. It is useful for scenario exploration and dashboard storytelling, but it should be refreshed and revalidated whenever the historical dataset or forecasting methodology changes.

Forecast horizon

Year	Forecast status
2027	Projected
2028	Projected
2029	Projected
2030	Projected

8. Key Business Insights

1. Long-term financial expansion

The year-wise financial trend shows a strong upward movement in aggregate financial amount across the historical period. This indicates substantial growth in the size of the measured financial structure over time.

2. Equity is the dominant category

Equity is the largest category in the project's aggregated composition, representing approximately 70.28% of the displayed financial composition. This makes equity structure an important area for management-level interpretation.

3. Financial composition changes over time

The category trend demonstrates that the mix of Assets, Borrowings, Equity, Liabilities and Other components varies by year. Monitoring composition is therefore more informative than relying on a single total.

4. Financial health is moderate on the project scale

The custom Financial Health Score is 50.08/100. The result should be used as a monitoring signal within this project, not as an external credit or investment rating.

5. Reserves and Surplus show a positive projected direction

The 2027–2030 forecast visualization shows continued upward movement in projected Reserves and Surplus, suggesting a positive modelled trajectory under the assumptions used.

6. Interactive analysis improves decision support

Year and Category slicers allow users to move from an executive overview to focused investigation without changing the underlying data preparation pipeline.

9. Business Recommendations

Strengthen year-wise monitoring

Use the dashboard as a periodic monitoring layer and compare new reporting years with the historical trend instead of reviewing isolated figures.

Monitor equity and liability structure

Because Equity dominates the project category composition, changes in equity and liability-related indicators should be monitored alongside overall financial growth.

Use category-level drill-down for management review

Investigate major movements by category before interpreting changes in the aggregate financial amount.

Treat the Financial Health Score as an internal indicator

Use the custom score for comparative dashboard monitoring and establish a documented methodology if it will be reused in future periods.

Refresh the forecast regularly

Re-estimate the 2027–2030 projection whenever new historical observations become available and compare forecast errors against actual results.

Maintain a controlled data pipeline

Keep the original source workbook unchanged and perform cleaning, modelling and reporting on controlled analytical copies. This reduces the risk of accidental alteration of source financial information.

Overall recommendation: the strongest long-term value of this solution is not the dashboard alone, but the repeatable pipeline connecting source data, Python preparation, SQL analysis, forecasting and Power BI reporting.

10. Analytical Workflow and Technology Stack

Stage	Implementation
Source data	Supplied TCS financial statement dataset
Preparation	Python / Pandas in Google Colab
Cleaning	Year standardisation, numeric conversion, indicator filtering, category classification
Quality control	Duplicate verification using Indicator + Year
Analysis	Descriptive statistics, cumulative component ranking, category trends
SQL layer	Structured financial analysis and analytical querying
Forecasting	Historical Reserves and Surplus projection for 2027–2030
BI modelling	Power BI data model and DAX measures
Reporting	Interactive Power BI executive dashboard

The workflow was deliberately modular: the source data was not overwritten, cleaning was performed on working copies, analytical outputs were exported for Power BI, and the dashboard was then validated using interactive slicer tests.

Tools used: Python, Pandas, Google Colab, SQL/SQLite, Microsoft Power BI and DAX.

11. Limitations and Data Governance

- The project report is based on the supplied dataset and the analytical outputs produced during the project. It does not independently audit the source financial statements.
- The Financial Health Score is a project-defined metric and should not be presented as an official TCS rating.
- The 2027–2030 forecast is model-based and should not be interpreted as management guidance.
- Category classification depends on the indicator-keyword logic used during cleaning; changes to that mapping may alter category totals.
- Some financial indicators contain zero values. These should be interpreted in their accounting context rather than automatically treated as missing data.
- Cumulative indicator rankings should not be confused with single-year financial statement totals.

Data governance principle: the original source dataset should remain immutable. Any future update should be added as a new version and passed through the same validation checks before the Power BI model is refreshed.

12. Conclusion

The TCS Corporate Financial Intelligence & Forecasting project demonstrates an end-to-end business analytics workflow that converts raw financial statement information into a structured decision-support system.

The completed solution combines data preparation, analytical validation, category intelligence, descriptive statistics, forecasting and interactive Power BI reporting. The final dashboard provides a concise executive view while retaining the ability to investigate the data through Year and Category filters.

The project's main analytical signals are a strong historical upward trend in aggregate financial amount, a dominant Equity component in the displayed category composition, a project-derived Financial Health Score of 50.08/100 and a positive projected direction for Reserves and Surplus through 2030.

For future iterations, the solution can be strengthened by incorporating additional financial statements, expanding the forecasting methodology, adding forecast-error tracking and documenting the exact formula and business thresholds for the Financial Health Score.

Final status: **Dashboard, analytical model, validation workflow and reporting layer completed.**

13. Data Sources and References

Primary project source

TCS financial statement / balance-sheet dataset supplied for this project. The project analysis, cleaning outputs and Power BI dashboard were developed from the supplied source dataset.

Official external reference

Tata Consultancy Services — Investor Relations / Financial Statements. The official TCS investor-relations portal provides annual reports, quarterly financial results and related financial disclosures.

Official investor-relations page: <https://www.tcs.com/investor-relations>

Official financial-statements page: <https://www.tcs.com/investor-relations/financial-statements>

A recent official TCS annual report was also used as a reference point for the existence and structure of TCS financial disclosures; project-specific dashboard values remain based on the supplied dataset and project calculations.

Annual Report reference:

<https://www.tcs.com/content/dam/tcs/investor-relations/financial-statements/2024-25/ar/annual-report-2024-2025.pdf>

Reproducibility note

For auditability, retain the original workbook, Google Colab notebook, exported analytical CSV/XLSX files, SQL scripts and final Power BI PBIX file together. Do not replace the original source dataset with cleaned data.

END OF REPORT